

Данное задание 2

$$(15, 7)_{\text{min}} = 5$$

1) порог. матрица в гв. виде.

2) декодир 111 000 000 11 000

$$t = \left\lfloor \frac{d_{\text{min}} - 1}{2} \right\rfloor = \left\lfloor \frac{5 - 1}{2} \right\rfloor = 2$$

$$15 = 2^4 - 1$$

$$\mathbb{GF}(2) \rightarrow \mathbb{GF}(2^4) \quad p(x) = 1 + x + x^4$$

$$\alpha^{15} = 1, \alpha - \text{примит. Эл.}$$

порогдательный многочлен имеет корни

$$\alpha^1 \quad \alpha^2 \quad \alpha^3 \quad \alpha^4$$

$$\alpha^0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0$$

$$\alpha^1 \quad x \quad 0 \quad 0 \quad 1 \quad 0$$

$$\alpha^2 \quad x^2 \quad 0 \quad 1 \quad 0 \quad 0$$

$$\alpha^3 \quad x^3 \quad 1 \quad 0 \quad 0 \quad 0$$

$$\alpha^4 \quad \dots$$

$$H = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$H = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$K = [A | I] \rightarrow G = [I | A^T]$$

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

- порогдательная матрица в двоичном виде

$$111001100011000$$

$$r(x) = x^{14} + x^{13} + x^{12} + x^9 + x^8 + x^4 + x^3$$

$$S_1 = r(\alpha) = \alpha^{12}$$

$$S_2 = r(\alpha^2) = \alpha^9$$

$$S_3 = r(\alpha^3) = \alpha^{14}$$

$$S_4 = r(\alpha^4) = \alpha^3$$

сиргравне $\neq 0 \Rightarrow$ есно аммбха

$$x = \alpha^i \quad y = \alpha^j$$

$$S_1 = x + y$$

$$S_2 = x^2 + y^2$$

$$S_3 = x^3 + y^3$$

$$z^2 + \sigma_1 z + \sigma_2 = 0$$

$$\sigma_1 = S_1 \Rightarrow \sigma_1 = \alpha^{12}$$

$$\sigma_2 = \frac{S_3 + S_1 S_2}{S_1}$$

$$\sigma_2 = \frac{\alpha^{14} + \alpha^{12} \cdot \alpha^9}{\alpha^{12}} = \frac{\alpha^{14} + \alpha^6}{\alpha^{12}} = \alpha^{-6} \neq \alpha^{11}$$

$$z^2 + \alpha^{12} z + \alpha^{11} = 0$$

$$(z+1)(z+\alpha^{11}) = 0 \Rightarrow z_1 = 1, z_2 = \alpha^{11}$$

аммбха $\Downarrow$ в нозмуре 0 u 11

$$\begin{array}{r} 111001100011000 \\ + 000100000000001 \\ \hline 111101100011001 \end{array}$$

$$C(x) = x^{14} + x^{13} + x^{12} + x^{11} + x^9 + x^8 + x^4 + x^3 + 1$$

$$C(x) = (x^6 + x^3 + 1) \cdot (x^8 + x^4 + x^6 + x^4 + 1)$$

$$m(x) = x^6 + x^3 + 1$$

" $y(x)$

$$\Downarrow$$

$$\Pi = \underline{1001001}$$